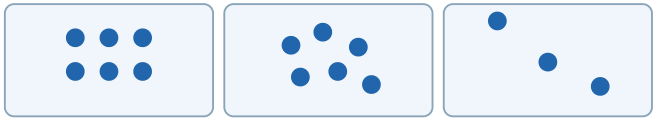
Particle model

Describe matter

1 States & density



Solid: vibrate in fixed positions. Liquid: close, move past. Gas: far apart, random movement.

$\rho = m/V$. Gas has low density from large spacing. Ice is less dense than liquid water.

2 Density practical

Balance → mass. Regular solid: dimensions → volume. Irregular solid: displaced water → volume.

Liquid: subtract empty-container mass. Read meniscus at eye level; avoid trapped bubbles.

Track internal energy

3 Temperature changes

Internal energy = total particle kinetic + potential energy. Temperature links to average kinetic energy.

$\Delta E = mc\Delta\theta$. c: energy to warm 1 kg by 1°C, J/kg°C.

4 State changes

Physical and reversible; mass conserved in a closed sample.

$E = mL$. During melting/boiling, potential energy rises, temperature stays constant. Fusion: solid/liquid; vaporisation: liquid/gas.

Explain pressure

5 Pressure from collisions

Gas particles hit walls → force per area. Force is perpendicular to wall.

At fixed volume and mass, warming → faster particles, more frequent/harder collisions → higher pressure.

6 Compress a gas

Fixed mass, constant temperature: $p_1V_1 = p_2V_2$. Smaller volume → more wall collisions → greater p.

Rapid compression does work on gas → internal energy and temperature can increase.